Quasi-Resonant Zero-Current-Switching Bidirectional Converter for Battery Equalization Applications

Yuang-Shung Lee, *Member, IEEE*, and Guo-Tian Cheng

***Abstract*—A systematic approach to reform and analyze a soft-switching bidirectional dc-to-dc converter is proposed for cell voltage balancing control in a series connected battery string. quasi-resonant converter circuits have been designed to achieve the zero-current-switching (ZCS) to reduce the switching loss in bidirectional battery equalizers. The results indicate that the switching loss and energy transfer efficiency can be substantially improved using the quasi-resonant ZCS (QRZCS) technology in a battery charging system with an individual cell equalizer (ICE). The validity of the battery equalization is further verified using an experimental installation involving a battery string of three lithium–ion cells. The simulation and experimental results show that the proposed QRZCS battery equalization schemes can achieve bidirectional battery equalization performance and reduce the MOSFET transistor switch power losses by more than 96% and increase the efficiency by around 20%∼30% compared with the conventional battery equalizer during an identical equalization process.**

***Index Terms*—Individual cell equalizer (ICE), quasi-resonant zero-current switching (QRZCS).**

I. Introduction

BATTERY cells connected in series are utilized in many practical applications such as electric vehicles (EVs), hybrid electric vehicles (HEVs), electric scooters (ES), electric wheelchairs, electric bikes, and telecom energy systems. Battery cell voltage imbalances are usually caused by the differences in cell residual capacities, internal resistance, degradation, and the ambient temperature gradient during charging or discharging. Imbalanced cell voltage in a battery string will give rise to overcharging or overdischarging, and decrease the total storage capacity and battery lifecycle [1], [2]. Therefore, maintaining cell voltage at an equal charge/discharge level is significant for enhancing the battery lifecycle. Cell voltage monitoring, current diverting battery equalization circuits, and battery management systems (BMS) for series connected battery strings have been presented in the literature to prevent cell voltage imbalances during charge and discharge periods in series connected battery strings [3]–[7].

Battery equalization control should be implemented to restrict the charge/discharge current to the allowable cell limitations in the battery string. This is determined by the cell design and its materials. A lithium–ion battery chemistry cannot withstand overcharge. An overcharge will vaporize the active materials in the battery, increase the internal pressure, and produce a high risk for explosion. The maximum overvoltage threshold is precariously close to the fully charged terminal voltage, typically in the range of 4.1∼4.2 V/cell [6]. During discharge, lithium–ion batteries must not be excessively discharged, i.e., down to the low voltage limit, typically about 2.0–2.5 V/cell [7]. Overdischarge will dissolve the copper in the electrolyte and form copper dendrites, which harms the battery and shortens the battery cell's lifecycle. Series connected lithium–ion batteries hide a more complex problem. Even though the pack voltage may appear to be within the acceptable limits, one cell in the series string may encounter damaging voltage due to cell-to-cell imbalances. Therefore, careful control and monitoring must be provided to prevent any single cell from experiencing over or under-voltage from excessive charging or discharging. Because a lithium–ion battery can not be overcharged and overdischarged, there is no natural gas mechanism to release the excessive energy to prevent cell voltage imbalance and damage. Therefore, an individual cell equalizer (ICE) must be employed to prevent, diagnose, and correct any cell voltage imbalance in a lithium–ion battery string [6], [7]. Cell balancing control is designed to obtain the maximum usable capacity from the lithium–ion battery string. Because battery string charging and discharging can be limited by any one single cell reaching its end-of-charge voltage threshold (about 4.1 to 4.2 V/cell) and low voltage threshold (about 2.0–2.5 V/cell), respectively, cell balancing algorithms seek to efficiently remove energy from a strong cell and transfer that energy into a weaker cell until the voltage is equalized to the same level across all cells. This enables additional charging capacity for the battery string [7]. A serious cell voltage imbalance is usually generated because of shrinkage in the internal impedance and/or cell capacity due to aging. Charged and discharged deviations of cells with diminished capacity and/or high internal impedance can produce large voltage swings. Additionally, in highly transient lithium–ion battery applications like regenerative EV or HEV, braking produces an inrush of instantaneous regenerative braking current that generates a sudden voltage increase in a lithium–ion battery, which in turn

Manuscript received October 21, 2004; revised July 15, 2005. This work was supported by the National Science Council of Taiwan, R.O.C. under Grant NSC 92-2213-E-030-020 and by the Ministry of Economic Affairs of Taiwan under Grant SBIR-1B940007. This work was presented in part at the 35th Annual IEEE Power Electronics Specialists Conference, Aachen, Germany, June 2004. Recommended by Associate Editor F. L. Luo.

Y.-S. Lee is with the Department of Electronic Engineering Fu-Jen Catholic University, Taipei 24205, Taiwan, R.O.C. (e-mail: lee@ee.fju.edu.tw).

G.-T. Cheng is with the Energy and Resource Laboratory, Industrial Technology Research Institute, Hsinchu 310, Taiwan, R.O.C.

Digital Object Identifier 10.1109/TPEL.2006.880349

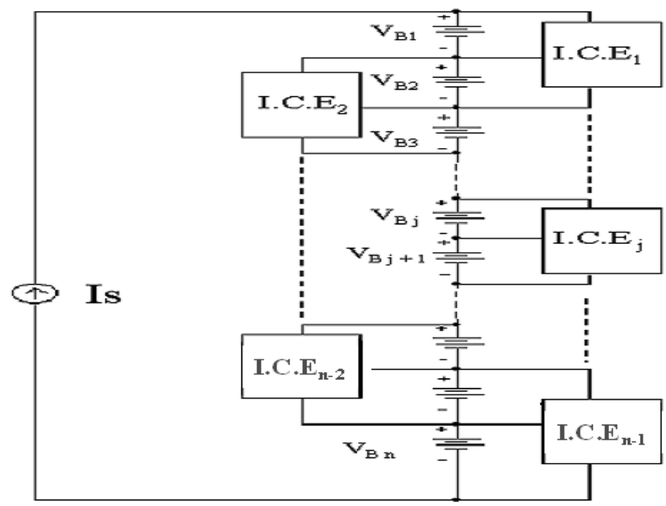

Fig. 1. Battery charging system with individual cell equalizers.

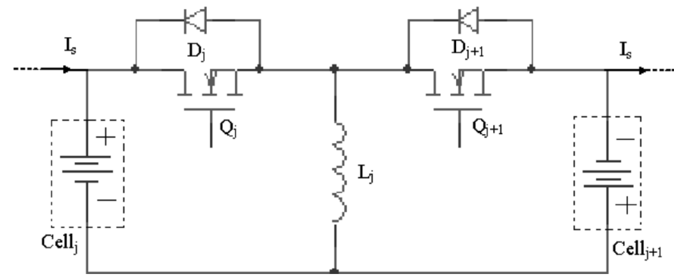

Fig. 2. Buck–boost battery equalizer.

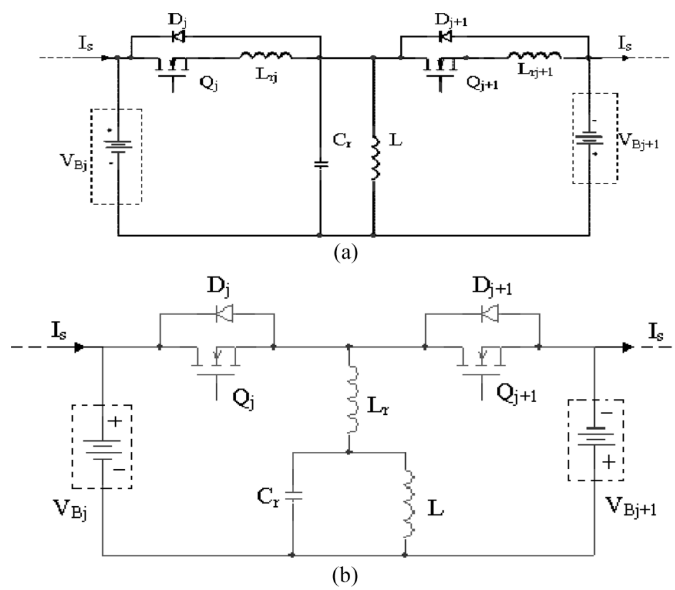

Fig. 3. Proposed QRZCS battery equalization schemes. (a) Buck–boost QRZCS battery equalizer. (b) Modified QRZCS battery equalizer.

creates an excessive electrolyte breakdown. Because of such overvoltage transient situations, it is necessary to balance the lithium cell chemistries [7]–[10].

This paper proposes a cell voltage balancing control scheme using a bidirectional dc-to-dc converter with quasi-resonant zero-current switching (QRZCS), which is basically a modified buck–boost converter. The cell voltages are controlled by the driving pulsewidth modulation (PWM) signals corresponding to the respective cell voltage, based on an equalization algorithm constructed in BMS microprocessors. The proposed battery equalization scheme, implementing QRZCS technology, can achieve zero-current soft switching, and reduce the MOSFET switching loss to increase the battery equalizer's efficiency. A design example for a series connected lithium–ion battery strings was conducted to achieve the predicted equalizing performance. The proposed QRZCS technology can reduce the equalization switching loss on the MOSFET and increase the equalization efficient of the battery equalization system. The electromagnetic interference (EMI) emissions into the battery charging source can also be suppressed due to the switching waveforms being along the sinusoidal function compared to a conventional battery string equalizing process during the charge state.

II. System and Topologies Description

Fig. 1 shows a battery charging system for a series connected battery string with individual cell equalizers (ICE). This study is particularly interested in an integrated infrastructure and modular design approach for the battery equalization scheme for practical battery pack applications. Fig. 2 shows a buck–boost bidirectional battery equalizer composed of MOSFET switches with body diodes and a large inductance inductor. This nondissipative equalization design has many advantages such as high equalization efficiency due to the nondissipative current diverter, bidirectional energy transferring capability, and a modular design [8]. The disadvantages of this equalization scheme are the hard switching of the bidirectional dc-to-dc converter, causing significant switching loss and the EMI emission in the battery charging system [9]–[12]. A simple zero-current soft switching technology based on the quasi-resonant converter theory can be used to mitigate the abovementioned drawbacks in reducing the battery equalizer switching loss and increasing the equalization efficiency, plus the advantage of low EMI emissions battery equalization system [13]–[15].

To achieve the quasi-resonant zero-current switching (QRZCS) effect, circuit elements, L_r and C_r must be added to construct a resonant tank circuit in the conventional buck–boost battery equalizer, as shown in Fig. 2. Fig. 3 shows the proposed ZCS ICE based on the quasi-resonant converter theory for the two alternating circuit topologies. Fig. 3(a) shows a buck–boost QRZCS battery equalizer. It is comprised of two resonant inductors L_{rj} and L_{rj+1}, a resonant capacitor C_r, a large energy storing inductance L, and two power MOSFETs with diodes employed as the cell balancing control switches. Assuming the main inductor L is adequately large, the current flowing through the inductor is nearly constant (I_L) [15]. The QRZCS effect can be achieved using the resonant tank circuit constructed by L_{rj} (L_{rj+1}) and C_r. Fig. 3(b) shows the modified QRZCS battery equalizer to refine from Fig. 3(a) with the minimal number of passive inductance elements and semiconductor switches under the same QRZCS equivalent topology. The energy transfer direction between the two adjacent cells is determined according to the cell voltage difference or the battery residual capacity (BRC) in the battery strings, controlled by the power MOSFET switches [10]. If $V_{Bj} > V_{Bj+1}$ and/or $\mathrm{BRC}_j > \mathrm{BRC}_{j+1}$, Q_j is turned on, and inductor L stores energy from V_{Bj} during the DT_S period. During the other $(1-D)T_S$ periods, Q_j is turned off, and the body diode D_{j+1} is forced to turn on. The stored energy in L is then discharged to V_{Bj+1} to act as a controlled charging current source for cell balance according to the energy transfer direction. If $V_{Bj} < V_{Bj+1}$ and/or $\mathrm{BRC}_j < \mathrm{BRC}_{j+1}$, Q_{j+1} is turned on during the DT_S period, and turned off during the other $(1-D)T_S$ period and the body diode D_j is forced to turn on, the stored energy in L is discharged into cell V_{Bj}, and the equalization energy is transferred to the reverse direction to balance the battery string adjoining cell voltage. The L_r and C_r are constructed as the resonant tank to achieve the QRZCS function for the symmetrical and bidirectional battery equalizer.

The buck–boost QRZCS battery equalizer is adopted and modified from the dc-to-dc QRZCS converter discussed in [14] and [15]. The following sections of this paper will focus on the modified QRZCS battery equalization scheme analysis.

III. Quasi-Resonant Battery Equalizers

The equivalent circuit in Fig. 3(b) for the proposed battery equalization scheme is depicted in Fig. 4(a) and (b) under the various cell voltages, $V_{Bj} > V_{Bj+1}$ and $V_{Bj} < V_{Bj+1}$, respectively. Some assumptions are necessary to analyze the steady state circuit behavior: 1) the main inductance L is sufficiently large; 2) the semiconductor switches are ideal; and 3) the tank circuit reactive elements are ideal. Therefore, the current flowing through the large inductance, L, is nearly constant, it can be represented as an equivalent current I_L that is controlled by the MOSFET switches. The direction of the controlled-current source is determined according to the desired cell balancing control for the battery string. The corresponding and typical switching waveforms (i_{Lr} and V_{cr}) of the modified QRZCS battery equalizer are also shown below in Fig. 4(a) and (b), respectively. The detailed equivalent circuits for the proposed battery equalization scheme are shown in Fig. 5(a)–(e) under different time intervals for $V_{Bj} > V_{Bj+1}$.

Stage 1 (Fig. 5(a); $t_0 < t < t_1$): The main switch Q_j is turned on at $t = t_0$, V_{Bj} provides current though MOSFET switch Q_j and the resonant tank L_r–C_r to store energy in the large inductor L. The initial voltage in C_r is $V_{\mathrm{C}}(t_0) = -V_0$, the dynamic equation in this interval is given by

$$L_r \frac{di_{Lr}(t)}{dt} + \frac{1}{C_r}\int (i_{Lr}(t) - I_L)\,dt = V_{Bj}. \tag{1}$$

Differentiating both sides of (1), a second-order differential equation is obtained as follows:

$$\frac{d^2 i_{Lr}(t)}{dt^2} + \omega_r^2 i_{Lr}(t) = \omega_r^2 I_L \tag{2}$$

with the initial conditions: $i_{Lr}(t_0) \cong 0$ and $di_{Lr}(t_0)/dt = ((V_{Bj} + V_0)/L_r)$. The solution for (2) can be obtained as

$$i_{Lr}(t) = \frac{V_{Bj} + V_0}{Z_r}\sin\omega_r t + (1 - \cos\omega_r t) I_L \tag{3}$$

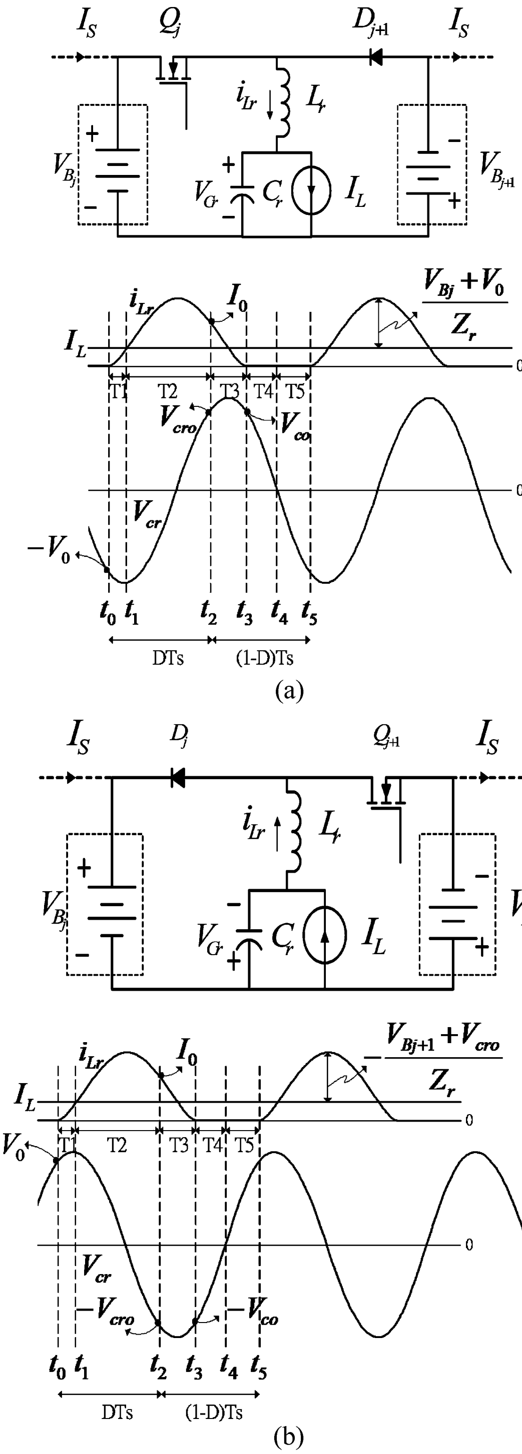

Fig. 4. Equivalent circuit and typical switching waveform of the modified QRZCS battery equalizer. (a) $V_{Bj} > V_{Bj+1}$. (b) $V_{Bj} < V_{Bj+1}$.

where the resonant angular frequency is $\omega_r = 1/\sqrt{L_r C_r}$, the normalized impedance is $Z_r = \sqrt{L_r/C_r}$. When $t = t_1$, $i_{Lr}(t_1) = I_L$, and let $t_0 = 0$ then the time t_1 is

$$T_1 = t_1 - t_0 = \frac{1}{\omega_r}\tan^{-1}\left(\frac{I_L Z_r}{V_{Bj} + V_0}\right). \tag{4}$$

Stage 2 (Fig. 5(b); $t_1 < t < t_2$): The main switch Q_j is still turned on in this interval. The inductor current will increase to

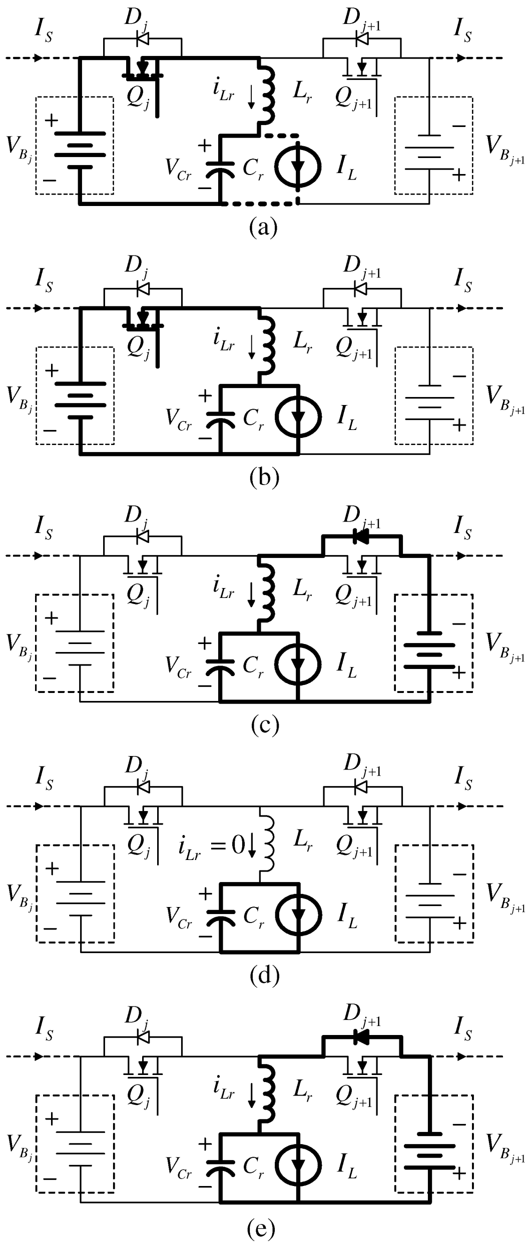

Fig. 5. Equivalent circuits for $V_{Bj} > V_{Bj+1}$ under various time intervals for the modified QRZCS battery equalizer. (a) $t = t_0 \sim t_1$. (b) $t = t_1 \sim t_2$. (c) $t = t_2 \sim t_3$. (d) $t = t_3 \sim t_4$. (e) $t = t_4 \sim t_5$.

reach the peak value, and substantially decrease until the main switch is turned off at $t = t_2$. The resonant inductor current is the same as that in (3) for $t_1 < t < t_2$. By using the terminal conditions $i_{Lr}(t_2) = I_0$, the time interval between t_1 and t_2 can be determined as

$$\begin{aligned} T_2 &= t_2 - t_1 \\ &= \frac{1}{\omega_r}\left[\tan^{-1}\left(\frac{I_L Z_r}{V_{B_j} + V_0}\right) - \sin^{-1}(I_L - I_0)\right] \\ &\quad - \frac{1}{\omega_r}\tan^{-1}\left(\frac{I_L Z_r}{V_{B_j} + V_0}\right) \\ &= \frac{-1}{\omega_r}\sin^{-1}(I_L - I_0). \end{aligned} \tag{5}$$

The resonant current flowing through the main switch is $I_{\mathrm{Lr}}(t)$, that is a half-cycle sinusoidal current with the amplitude $((V_{Bj} + V_0)/Z_r)$ and is based on the above of dc leveling current I_L. The conditions of the maximal resonant inductor current for obtaining the ZCS is $I_{Lr\,\max} \le ((V_{Bj} + V_0)/Z_r)$ shown as in Fig. 4(a), and the peak current of the I_L is $I_L + ((V_{Bj} + V_0)/Z_r)$. The terminal condition of the resonant inductor current $i_{Lr}(t_2) = I_0$, which can be chosen between the peak current of the main switch and the equivalent current I_L for achieving the effect of ZCS. The condition has been derived from literature [15] and is expressed as

$$I_L \le I_0 \le I_L + \frac{V_{Bj} + V_0}{Z_r}. \tag{6}$$

Stage 3 (Fig. 5(c); $t_2 < t < t_3$): When Q_j is turned off at $t = t_2$, the resonant inductor current simultaneously switches to the body diode D_{j+1} which is forced to turn on and charges the cell V_{Bj+1} through the L_r–C_r resonant tank circuit. The dynamic equation in this interval can be written as

$$L_r \frac{di_{Lr}(t)}{dt} + \frac{1}{C_r}\int (i_{Lr}(t) - I_L)\,dt = -V_{Bj+1} \tag{7}$$

where $i_{Lr}(t_2) = I_0$, $di_{Lr}(t_2)/dt = -((V_{Bj+1} + V_{cro})/L_r)$, and where $V_{cr}(t_2) = V_{cro}$ are the initial conditions of the differential equation. The solution for (7) can be obtained as

$$\begin{aligned} i_{Lr}(t) &= (I_0 - I_L)\cos[\omega_r(t - t_2)] \\ &\quad - \frac{V_{Bj+1} + V_{cro}}{Z_r}\sin[\omega_r(t - t_2)] + I_L. \end{aligned} \tag{8}$$

The inductor current will decrease along the sinusoidal function to I_L and continuously fall to zero at $t = t_3$. Therefore, the time T_3 can be solved by $i_{Lr}(t_3) = 0$ as

$$T_3 = t_3 - t_2 = \frac{1}{\omega_r}\left\{\tan^{-1}\left[\frac{Z_r \times (I_0 - I_L)}{V_{B_{j+1}} + V_{cro}}\right] + \sin^{-1} I_L.\right\} \tag{9}$$

As a result, the voltage across the capacitor C_r alternates with the sinusoidal function. When $t = t_3$, the corresponding value V_{co} of the capacitor voltage $V_c(t_3)$ is

$$\begin{aligned} V_{co} &= (I_0 - I_L) Z_r \sin\omega_r(t_3 - t_2) \\ &\quad + (V_{Bj+1} + V_{cro})\cos\omega_r(t_3 - t_2) - V_{Bj+1}. \end{aligned} \tag{10}$$

Stage 4 (Fig. 5(d); $t_3 < t < t_4$): In this interval, Q_j and D_{j+1} are both turned off, the inductor current $i_{Lr}(t) = 0$, the capacitor is discharged to the constant current source by I_L, and the voltage across the capacitor C_r is

$$V_{cr}(t) = V_{co} - \frac{1}{C_r}\int I_L dt. \tag{11}$$

When $t = t_4$, the voltage $V_{cr}(t)$ decrease linearly from V_{co} to zero. The time T_4 can be determined from (11) using the terminal condition as follows:

$$\begin{aligned} T_4 &= t_4 - t_3 \\ &= \frac{V_{co} C_r}{I_L} - \frac{1}{\omega_r} \\ &\quad \times \left\{\tan^{-1}\left[\frac{Z_r \times (I_o - I_L)}{V_{B_{j+1}} + V_{cro}}\right] + \tan^{-1}\left(\frac{I_L Z_r}{V_{B_j} + V_0}\right)\right. \\ &\quad \left. + \sin^{-1} I_L - \sin^{-1}(I_L - I_0)\right\}. \end{aligned} \tag{12}$$

Stage 5 (Fig. 5(e); $t_4 < t < t_5$): This is a free-wheeling stage through the main inductor L, cell voltage V_{Bj+1} and body diode

D_{j+1}, when the capacitor current $i_{cr}(t)$ is smaller than I_L, the free-wheeling diode D_{j+1} is forced to turn on. The equivalent circuit and inductor current are identical to that in Stage 3. Therefore, the sinusoidal voltage $V_{cr}(t)$ is substantially decreased to reach the negative peak. It then increases until the next switching cycle. The free wheeling current is very small during this interval, so the $i_{Lr}(t)$ can be neglected for simplified analysis. The time for this period can be approximately found using

$$\begin{aligned} T_5 &= t_5 - t_4 \\ &= \pi - \frac{1}{\omega_r}\left[\tan^{-1}\left(\frac{I_L Z_r}{V_{B_j}+V_0}\right) - \sin^{-1} I_L\right] + \frac{V_{co}C_r}{I_L}. \end{aligned} \tag{13}$$

For the given parameters and initial conditions of the proposed modified QRZCS battery equalizer, the period of the entire cycle of the converter can be determine by

$$T_S = T_1 + T_2 + T_3 + T_4 + T_5. \tag{14}$$

The switching frequency is

$$f_S = \frac{1}{T_S}. \tag{15}$$

And the conduction duty ratio is

$$D = \frac{T_1 + T_2}{T_S}. \tag{16}$$

Assume the active and passive devices in the modified QRZCS battery equalizer are ideal and the inductor current in the large inductor L is continuous during the steady state. Fig. 6(a) and (b) show the equivalent circuit containing inductor L during interval $t_0 \sim t_2$ and $t_2 \sim t_5$, respectively. The inductor current $i_L(t)$ is

$$\begin{aligned} i_L(t) &= \begin{cases} -\left(\frac{V_{Bj}}{L_r+L} + \frac{V_0}{L}\right) \times \frac{1}{\omega_e} \sin\omega_e(t-t_0) \\ \quad + \frac{V_{Bj}}{L_r+L}(t-t_0), & \text{for } t_0 \le t \le t_2 \\ \left(\frac{V_{Bj+1}}{L_r+L} + \frac{V_{cro}}{L}\right) \times \frac{1}{\omega_e} \sin\omega_e(t-t_2) \\ \quad - \frac{V_{Bj+1}}{L_r+L}(t-t_2), & \text{for } t_2 \le t \le t_5 \end{cases} \\ &= \begin{cases} I_{\mathrm{Lm}} \sin\omega_e(t-t_0) + \frac{V_{Bj}}{L_r+L}(t-t_0), & \text{for } t_0 \le t \le t_2 \\ I_{Lm^*} \sin\omega_e(t-t_2) - \frac{V_{Bj+1}}{L_r+L}(t-t_2), & \text{for } t_2 \le t \le t_5 \end{cases} \end{aligned} \tag{17}$$

where the angular frequency $\omega_e = 1/\sqrt{L_e C_r} = \sqrt{(L_r+L)/LL_rC_r}$ and the equivalent inductance $L_e = L// L_r = (LL_r/(L_r+L))$.

The average inductor current I_L is

$$\begin{aligned} I_L = \frac{1}{\omega_e^2 T_S} &\times \left\{\left(\frac{V_{Bj}}{L_r+L} + \frac{V_0}{L}\right) \times (\cos\omega_e DT_S - 1)\right. \\ &\left. - \left(\frac{V_{Bj+1}}{L_r+L} + \frac{V_{cro}}{L}\right) \times [\cos\omega_e(1-D)T_S - 1]\right\} \\ &+ \frac{T_S}{2(L_r+L)} \times \left[D^2 V_{Bj} - (1-D)^2 V_{Bj+1}\right]. \end{aligned} \tag{18}$$

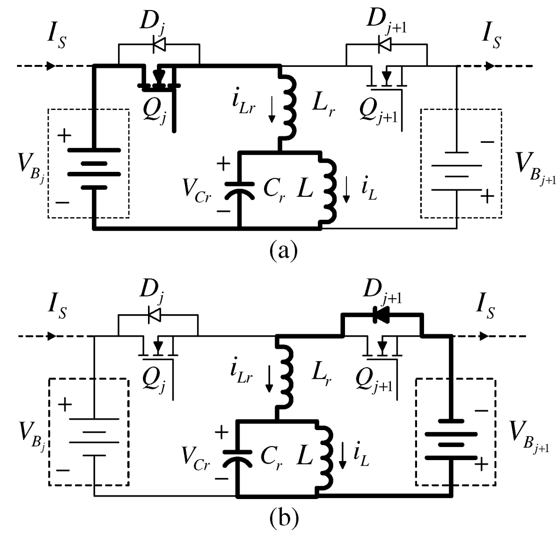

Fig. 6. Equivalent circuits containing inductor L of the modified QRZCS battery equalizer for $V_{Bj} > V_{Bj+1}$ under various time intervals. (a) $t = t_0 \sim t_2$. (b) $t = t_2 \sim t_5$.

The deviation of the inductor current is Δi_L which can be predicted by the following approximation

$$\begin{aligned} \Delta i_L &= \frac{1}{2}\{I_{Lm} + I_{Lm^*}\} \\ &= \frac{1}{2}\left\{-\left[\frac{V_{Bj}}{\omega_e(L_r+L)} + \frac{V_0}{\omega_e L}\right] + \frac{V_{Bj+1}}{\omega_e(L_r+L)} + \frac{V_{cro}}{\omega_e L}\right\} \\ &= \frac{1}{2\omega_e}\left\{\frac{V_{Bj+1} - V_{Bj}}{(L_r+L)} + \frac{V_{cro} - V_0}{L}\right\}. \end{aligned} \tag{19}$$

The average current and deviation current in the large inductor L can be predicted by using the approximation (18) and (19), respectively. The small deviation of the inductor current Δi_L can be obtained by choosing a larger L and/or suitable switching period.

Fig. 4(b) shows the alternating equivalent circuit and the typical switching waveform of the bidirectional battery equalizer for $V_{Bj} < V_{Bj+1}$. The working principle and analytical procedure are similar to that for the bidirectional battery equalizer for $V_{Bj} > V_{Bj+1}$, and are not discussed in this paper.

IV. Simulation and Experimental Verifications

To verify the analysis results discussed above, a PSpice simulation was performed for a battery stack of three battery cells with two proposed QRZCS equalization schemes. Fig. 7 shows the simulation setup for the proposed battery equalization schemes. Fig. 7(a) illustrates the system configuration for the series connected battery string composed of a three cell batteries with two proposed ICEs, and a battery management system (BMS) including a sensor monitoring the battery state, and the equalization controller. A charging source, I_S, is employed to simulate the following states of the battery pack: $I_S > 0$ denotes that the system is in charging state, $I_S < 0$ denotes that the system is in discharging state, and $I_S = 0$ represents the floating state that is neither charging nor discharging takes place. Fig. 7(b) and (c) shows the jth ICE for the buck–boost QRZCS and modified QRZCS battery equalizers, respectively.

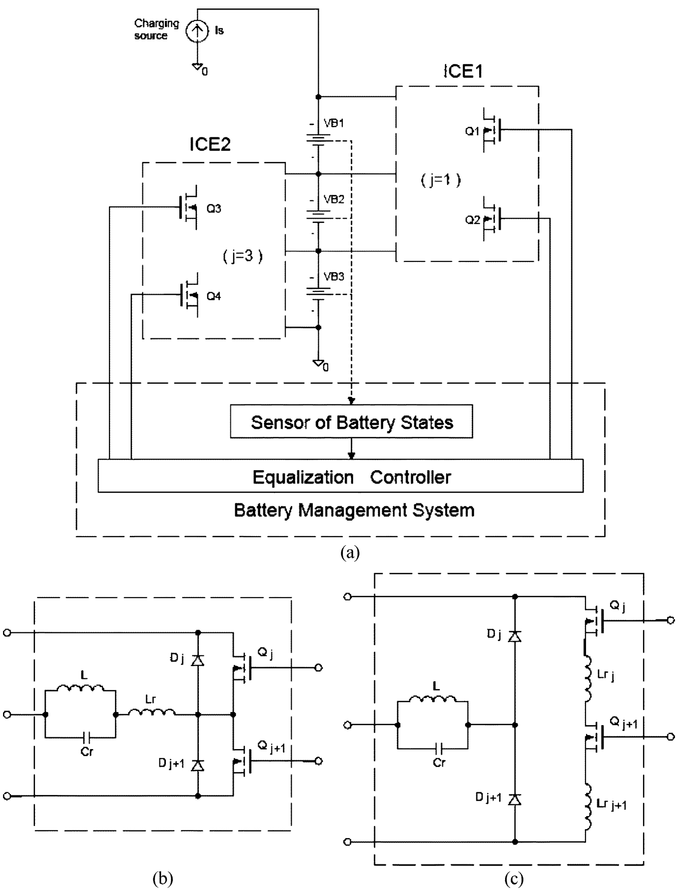

Fig. 7. Simulation setup for the proposed battery equalization schemes. (a) System configuration. (b) Modified QRZCS battery equalizer. (c) Buck–boost QRZCS battery equalizer.

The simulation parameters are the same as tabulated in the Table I. The BMS handles the cell balancing control of the battery string according to the characteristics and the operation limits of the lithium–ion battery pack. The strategy and experience for the cell equalization control of a lithium–ion battery string are summarized as follows.

- The equalization algorithm will be started when the voltage difference between two adjoining cells exceeds 0.0196 (V) (hardware resolution limit of A/D conveter) to minimize the cell-to-cell imbalance.
- During the charging equalization state, the cell voltage can not exceed its end-of-charge voltage (about 4.1 V/cell) to prevent overcharging and dissolving the copper in electrolyte which would damage the active materials of the lithium–ion cell.
- During the discharging equalization state, the cell voltage can not go below its low voltage threshold (about 2.8 V/cell) to prevent overdischarge and damaging the cell capacity and its life.
- If either one cell voltage in the battery string exceeds its end-of-charge voltage during charge equalization state, or one cell voltage in the battery string reaches its low voltage threshold during discharge equalization, the BMS will send a command to stop the cell voltage balancing process.

TABLE I
COMPARISON OF EQUALIZER WITH AND WITHOUT QRZCS

	Buck-boost equalizer	Modified QRZCS equalizer	Buck-boost QRZCS equalizer
Designed parameters	L=273uH	L=273uH L_r=59.2uH C_r=1.66uF	L=273uH $L_{rj}=L_{rj+1}$=59.2uH C_r=1.66uF
Components	2 MOSFETs 1 Inductor	2 MOSFETs 2 Inductors 1 Capacitor	2 MOSFET$_s$ 2 Diodes 3 Inductors 1 Capacitor
Switching parameters	D=0.5 f_s=16.67kHz	D=0.5 f_s=16.67kHz	D=0.5 f_s=16.67kHz
Average inductor current I_L	1.37A	1.07A	0.95A
Soft switching	No soft switching	Soft turn-off	Soft turn-on Soft turn-off
Switch peak voltage stress	$V_{Bj}+V_{Bj+1}$	$V_{Bj}+V_{Bj+1}$	$V_{Bj}+V_{Bj+1}$
Switch peak current stress	Eq. (13)	Eq. (13)	Eq. (13)
Average equalization efficiency	η = 52 % (η_{max} =53 %)	η = 79.4 % (η_{max} =83.1 %)	η = 80 % (η_{max} =85.6 %)

For simplicity reasons, the battery storage elements were assumed to be capacitors established in the PSpice simulation package library using a capacitor with an equivalent series resistor (ESR). In practical simulations, the capacitance is chosen smaller than the theoretical value to stabilize and speed up the simulation process. The battery initial voltages in the simulations were set as $V_{\rm B1} = 4.0$ V, $V_{\rm B2} = 3.5$ V, and $V_{\rm B3} = 3$ V. The capacitance of the represented battery $C_{\rm bat}$ was selected as 5 mF with $\mathrm{ESR} = 0.01\ \Omega$. The circuit parameters of the modified QRZCS battery equalizer were $L = 270\ \mu$H, $L_r = 59.6\ \mu$H and $C_r = 1.66\ \mu$F, shown as in Fig. 7(b). The switching frequency for the equalization converter was chosen by (15) (about 16.66 kHz), and the duty ratio was determined by (16) (about $D = 0.5$) for both $V_{\rm B1} > V_{\rm B2} > V_{\rm B3}$ and $V_{\rm B1} < V_{\rm B2} < V_{\rm B3}$. The detail simulation parameters are shown in the third column of Table I. The proposed battery equalizers were designed to operate at the zero-current-switching mode that can obtain a lower switching loss to improve the cell voltage balancing scheme efficiency.

Fig. 8(a) and (c) show the simulation results for the proposed battery equalizer under $V_{\rm B1} > V_{\rm B2} > V_{\rm B3}$. Fig. 8(a) shows the resonant capacitor voltage and inductor currents, respectively. Fig. 8(c) shows the cell voltage trajectories during cell voltage equalization under floating state. One can clearly observe that the resonant inductor current $I_{\rm Lr}(t)$ has sinusoidal waveform and swing in the positive half cycle from zero through the peak value returning to zero. The large inductor current I_L is nearly constant and is the same as in the mentioned description. The corresponding capacitor voltage has also sinusoidal waveform in a nonsymmetrical full cycle. The peak value will occur at the quarter period, different from the inductor current. The main switches are turned on at near zero-current and turned off at zero-current which validates that the converter operates in QRZCS to reduce the MOSFET switching loss. Fig. 9(a) and (c) show the alternating simulation results for the same battery equalizer under $V_{\rm B1} < V_{\rm B2} < V_{\rm B3}$. For convenient comparison, Fig. 11 shows the same simulations

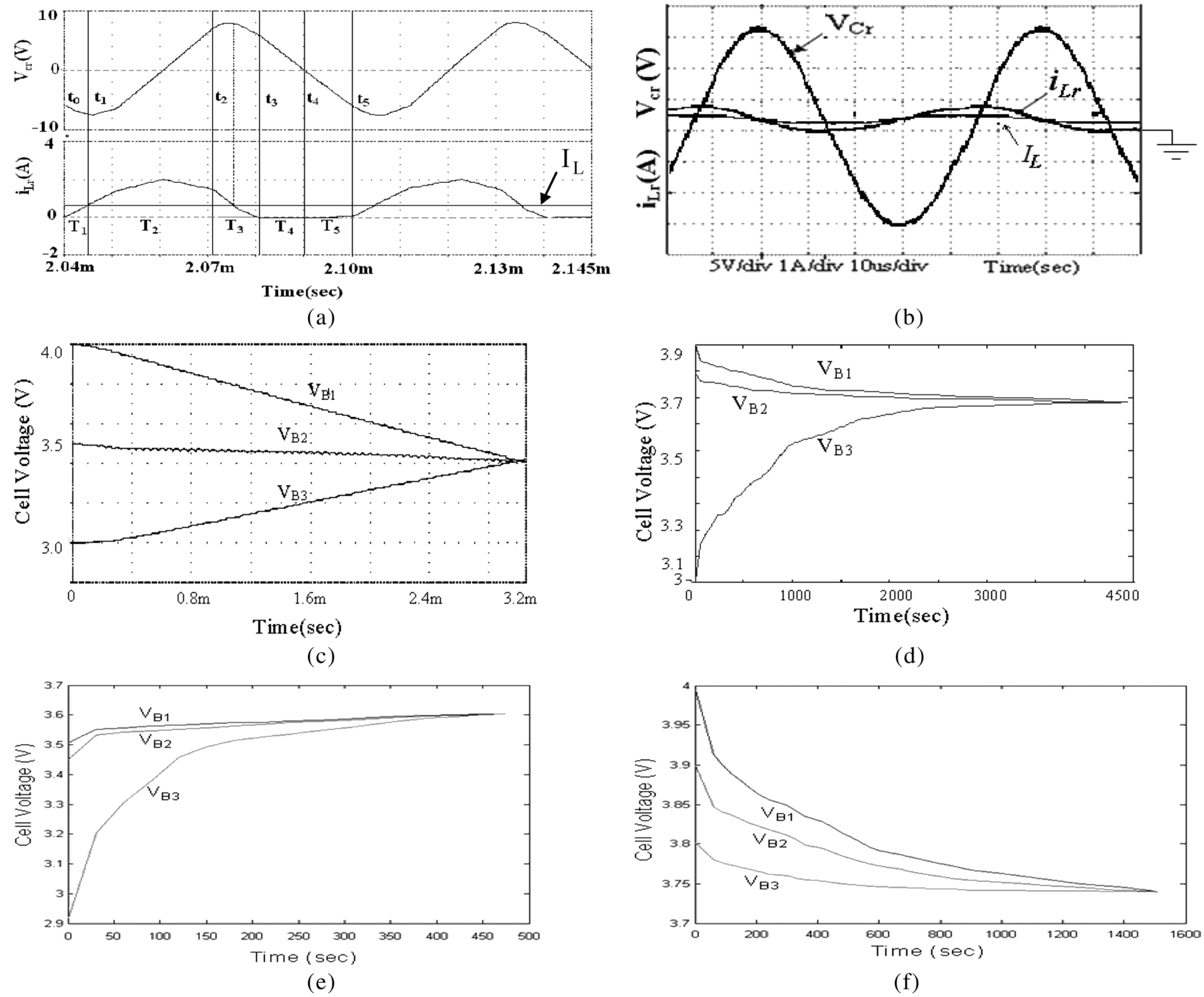

Fig. 8. Simulations and experimental results for $V_{B1} > V_{B2} > V_{B3}$ for the modified QRZCS battery equalizer. (a) Simulation waveform of V_{cr}, I_L, and i_{Lr}. (b) Experimental waveform of V_{cr}, I_L, and i_{Lr}. (c) Simulation cell voltage trajectories under floating state. (d) Experimental cell voltage trajectories under floating state. (e) Experimental cell voltage trajectories under added charging current 1 A. (f) Experimental cell voltage trajectories under added discharging current 1 A.

and experimental results for the battery string equipped with the proposed buck–boost QRZCS battery equalizer with parameters $L_{rj} = \mathrm{Lr}_{j+1} = 59.2\ \mu$H, $L = 273\ \mu$H, and $C_r = 1.66\ \mu$F for the identical converter switching conditions, shown as in Fig. 7(c). The simulation parameters are tabulated in the forth column of Table I. Fig. 11(a) and (c) show the simulation results for the buck–boost QRZCS battery equalizer under $V_{\mathrm{B1}} > V_{\mathrm{B2}} > V_{\mathrm{B3}}$. Fig. 11(a) shows the resonant capacitor voltage and inductor current of the buck–boost QRZCS battery equalizer. Fig. 11(c) shows the cell voltage trajectories during cell voltage equalization for the same equalizer under floating state. Observation from Figs. 8(c), 9(c), and 11(c), after the all cell voltages are balanced within the cell voltage sensing device limitations, the BMS will stop the battery equalizer cell voltage balance processing, therefore, all cell voltages are halted at the same end-of-charge state for the battery string.

To validate the proposed equalization scheme, a battery stack with two of the proposed modular equalization schemes was installed in three-cell battery strings to verify the QRZCS battery equalization scheme performance, as shown in Fig. 12. The driving signals for the two equalization schemes were controlled using a microprocessor based BMS according to the measured cell voltages. The driving signals were constructed using a logical switching algorithm that was generated from the BMS, and by using an 8052 microprocessor. The battery stacks were MRL/ITRI-10Ah lithium–ion battery cells with the initial cell voltages $V_{\mathrm{B1}} = 3.9$ V, $V_{\mathrm{B2}} = 3.796$ V, $V_{\mathrm{B3}} = 3.004$ V. The inductance and capacitance were, $L = 273\ \mu$H, $L_r = 59.2\ \mu$H and $C_r = 1.66\ \mu$F, respectively. The battery equalization system was driven using a nominal switching frequency of 16.67 kHz with the mentioned duty ratio for the ICE switching devices IRF530. Fig. 8(b), (d) and 9(b), (d) show the experimental resonant inductor currents, I_{Lr} and I_L, capacitor voltage V_{cr}, and the three-cell voltage trajectory results during equalization in the floating state for the modified QRZCS battery equalization scheme, respectively. Even though the

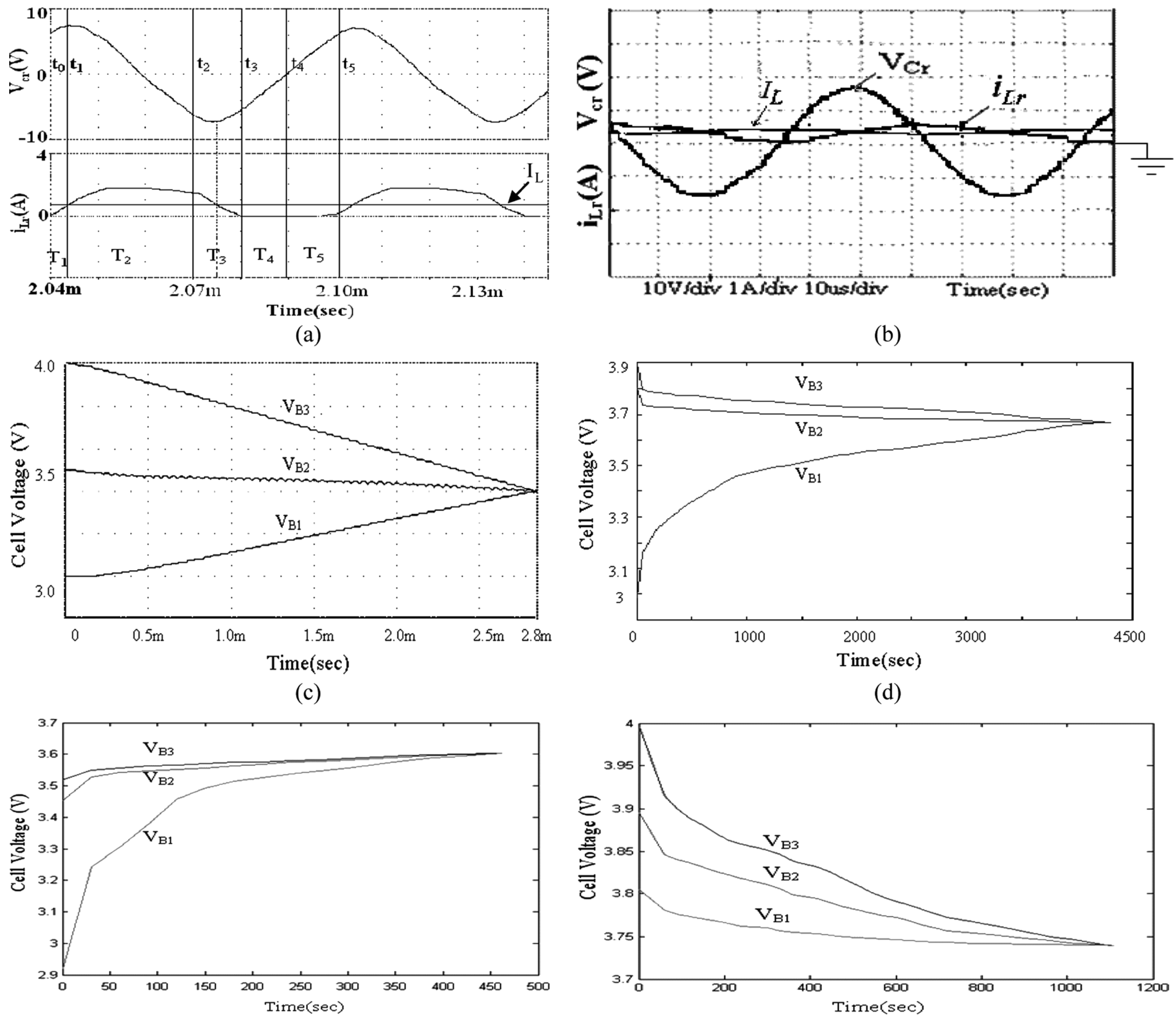

Fig. 9. Simulations and experimental results for $V_{B1} < V_{B2} < V_{B3}$ for the modified QRZCS battery equalizer. (a) Simulation waveform of V_{cr}, I_L, and i_{Lr}. (b) Experimental waveform of V_{cr}, I_L, and i_{Lr}. (c) Simulation cell voltage trajectories under floating state. (d) Experimental cell voltage trajectories under floating state. (e) Experimental cell voltage trajectories under added charging current 1 A. (f) Experimental cell voltage trajectories under added discharging current 1 A.

inductor current (I_L) has a small deviation in the experimental measurement, it can be improved by selecting a larger inductance for the study. Fig. 10(a) and (b) show the experimental results of $i_L(t)$ and $i_{Lr}(t)$ under various inductances L. The deviation current ratio of $\Delta i_L/I_L$, $\Delta i_{Lr}/I_{Lr}$ and $\Delta i_L/\Delta i_{Lr}$ are plotted in Fig. 10(c) for various inductances L. We can see that the $(\Delta i_{Lr}/I_{Lr}) > 1$ for various inductances L, it shows that the resonant inductor current is operated at discontinuous conduction mode. The per-unit deviation currents $\Delta i_L/I_L$ and $\Delta i_{Lr}/I_{Lr}$ are decreased for increasing inductance L. The measurements are the same as the results of the mentioned theoretical analysis shown in (17) and (18). Figs. 8(e), 9(e), 8(f), and 9(f) show the same cell voltage trajectories under added a 1-A charging and discharging current, respectively. Fig. 11(b), (d)–(f) show the resonant inductor currents, i_{Lr} and capacitor voltage V_{cr}, and the three-cell voltage trajectory experimental results during equalization in the same three operating states for the battery string with the buck–boost QRZCS battery equalizer, respectively. The experimental results are identical to those shown in the mentioned theoretical analysis and simulations. This obviously demonstrates that the proposed ICEs are designed to perform cell voltage balancing control under zero-current switching for the bidirectional battery equalizer. After all cell voltages are balanced within the resolution limits of the analog-to-digital converter, the BMS will send an execution command to cut-off the MOSFETs, and stop the cell voltage balance processing for the battery equalizer. Therefore, the equalization method can balance all adjoining battery string cell voltages to the same voltage level. Consequently, each cell can be simultaneously charged or discharged to its end-of-charge voltage. The total battery string charging/discharing capacity will be increased. Because of the

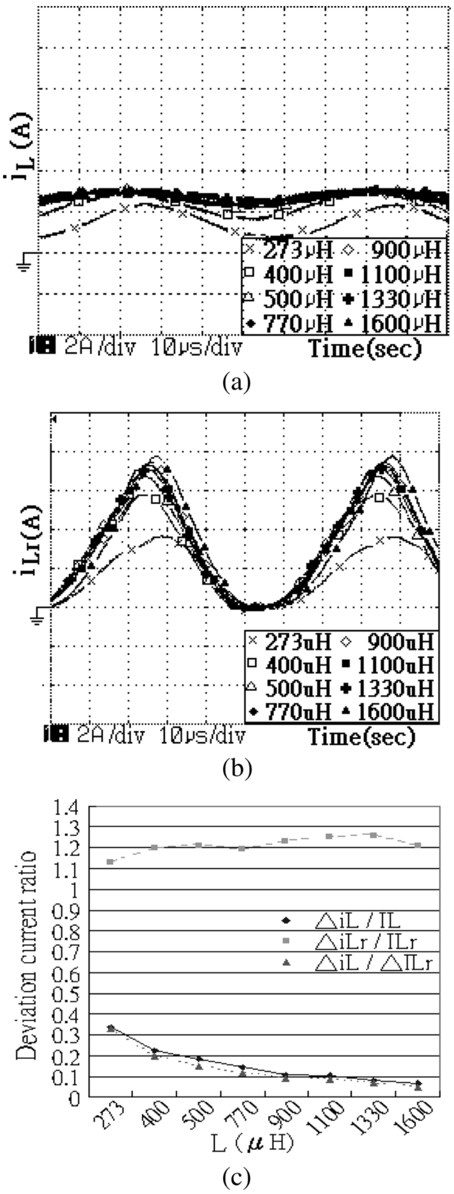

Fig. 10. Experimental results of i_{Lr} and i_L under various inductances L. (a) Experimental waveform of i_L. (b) Experimental waveform of i_{Lr}. (c) Experimental deviation current ratio of i_{Lr} and i_L.

switching loss in the MOSFET parasitic capacitors is smaller under the selected switching frequency, it is neglected in this study. Fig. 13(a)–(c) plot the experimental MOSFET switch power losses $P_{\mathrm{T}} = \nu_{\mathrm{T}}\, I_{\mathrm{T}}$ results for the bidirectional battery equalizer during the equalizing period under hard switching and the proposed soft switching, respectively. From the measured results P_{T} of Fig. 13(b) and (c), it is clear that the proposed equalization scheme with the designed QRZCS can reduce the power losses in the MOSFET switch during turn-on and turn-off by more than 96% compared with the same equalization scheme without ZCS. Fig. 13(d) shows the FFT frequency spectrum amplitude of the measured power losses for the proposed battery equalizer without and with QRZCS technology, respectively. This clearly shows that the dc component and low-order harmonics are high in the system without QRZCS. However, the high-order harmonic components attenuate very quickly. The FFT spectrum amplitude is significantly reduced in the system with the two proposed QRZCS schemes. Fig. 14 shows the measured results of the converter efficiency with respect to the ICE transfer energy under the same specified cell voltage equalizing process for the buck–boost, buck–boost QRZCS and modified QRZCS battery equalizers, respectively. From the analyzed spectrum results, the EMI emission for the proposed battery equalizer with QRZCS complies better with the EMC regulations than for the battery equalizer without QRZCS. The energy transfer efficiency of the two proposed QRZCS battery equalizers is high at about 20%∼30% compared with the conventional buck–boost battery equalizer. The maximum efficiency of the battery equalizer can be improved from 53% to reach around 83%∼85%.

V. Comparison of Equalizer With and Without QRZCS

In order to obtain a more complete comparison about the use of bidirectional converters with and without QRZCS as battery equalizer, a detailed designed results and performance will be further evaluated based on the same converter switching conditions and the circuit parameters in the Table I. Due to the number of converter components in the buck–boost QRZCS equalizer being more than in the other two equalizers, the costs for this design are the highest and the lower equalizing current will increase the equalization time during the equalizing process. Because of the buck–boost QRZCS battery equalizer can be designed to operate at zero-current turn-on and zero-current turn-off, the equalization efficiency is slightly better than the modified QRZCS equalizer. The equations for the peak voltage stresses of the switches for the three types of battery equalizers are shown in Table I. Observations from Figs. 8–11, Figs. 13, 14, and Table I, several features of the proposed QRZCS battery equalizers are summarized and shown as follows.

- The peak current stress of the jth switch in the three type battery equalizers are [16]

$$I_{pTj} = \begin{cases} I_{Bj+1} + \frac{V_{Bj} T_S}{L} & \text{Buck-boost equalizer} \\ I_L\left(1 + \frac{I_L Z_r}{V_{Bj} + V_o}\right) & \\ \quad + \frac{V_{Bj} + V_o}{Z_r} & \text{Modified QRZCS equalizer} \\ I_{Bj} + I_{Bj+1} & \\ \quad + \frac{V_{Bj} + V_{Bj+1}}{Z_r} & \text{Buck-boost QRZCS equalizer.} \end{cases} \tag{20}$$

- The power MOSFETs of the modified QRZCS battery equalizer are turned off at the zero-current state, but turned on at a small current state. The total switching losses of the MOSFETs in the proposed QRZCS battery equalizers are significantly reduced.
- The converter efficiency of the two proposed QRZCS battery equalizers are significantly improved compared with the conventional buck–boost battery equalizer. The energy transfer efficiency is increased 20%∼30% during the same cell voltage equalization process.
- The switching waveforms in the converters match the sinusoidal function and, therefore, the EMI emissions in the battery charging source can be suppressed.
- The number of inductor elements and diodes in the modified QRZCS battery equalizer are reduced compared with

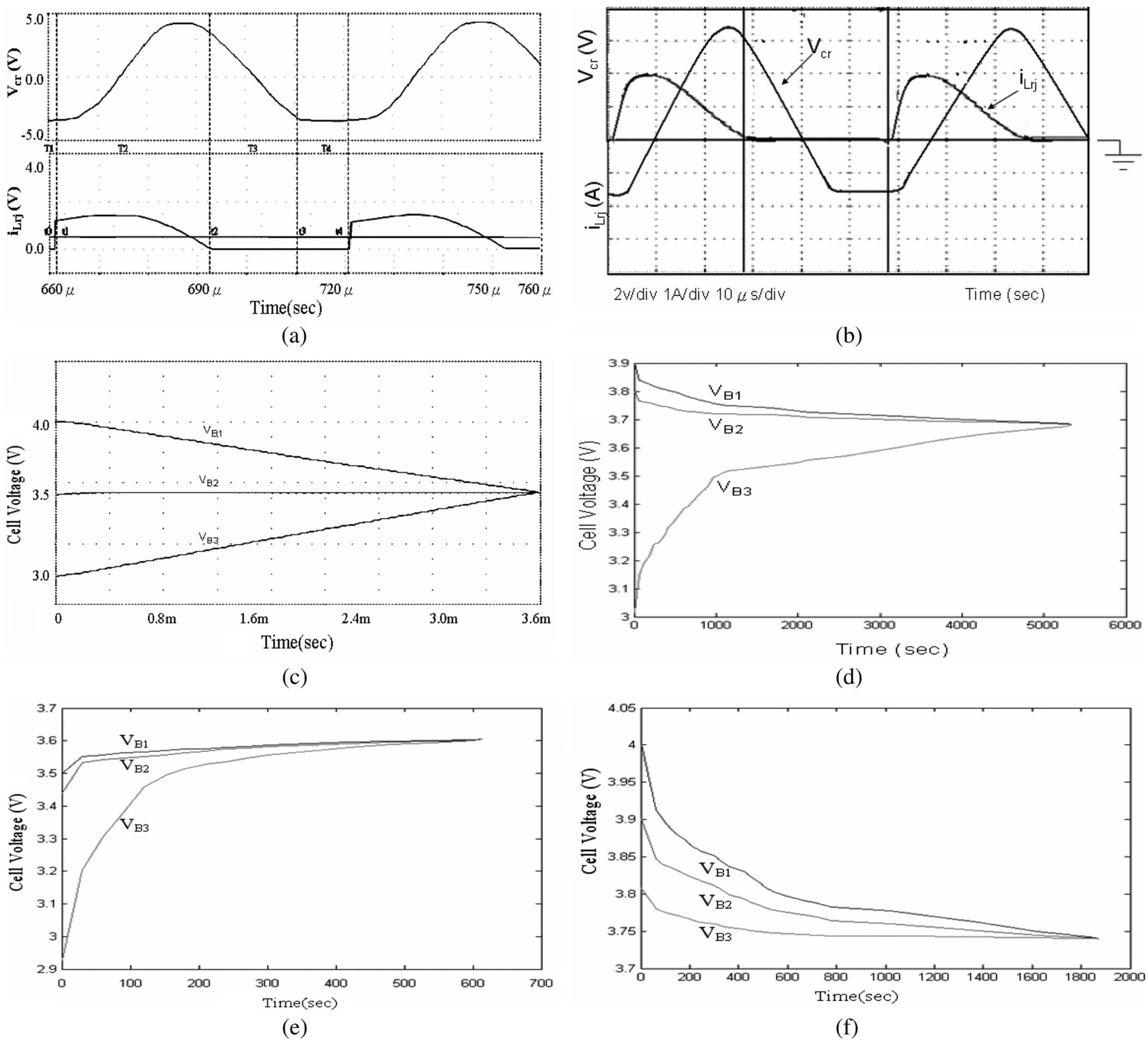

Fig. 11. Simulations and experimental results for $V_{B1} > V_{B2} > V_{B3}$ for the buck–boost QRZCS battery equalizer. (a) Simulation waveform of V_{cr} and i_{Lrj}. (b) Experimental waveform of V_{cr} and i_{Lrj}. (c) Simulation cell voltage trajectories under floating state. (d) Experimental cell voltage trajectories under floating state. (e) Experimental cell voltage trajectories under added charging current 1 A. (f) Experimental cell voltage trajectories under added discharging current 1 A.

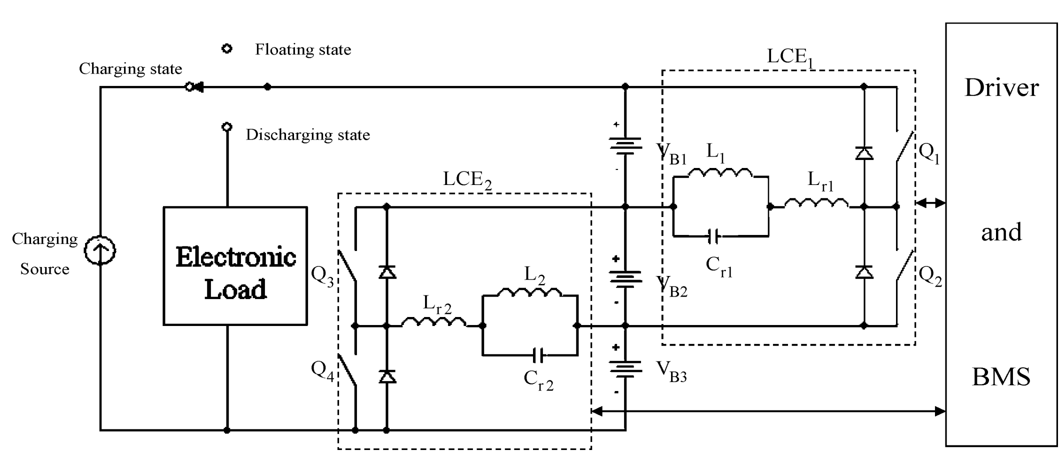

Fig. 12. Experimental installation of modified QRZCS battery equalizer.

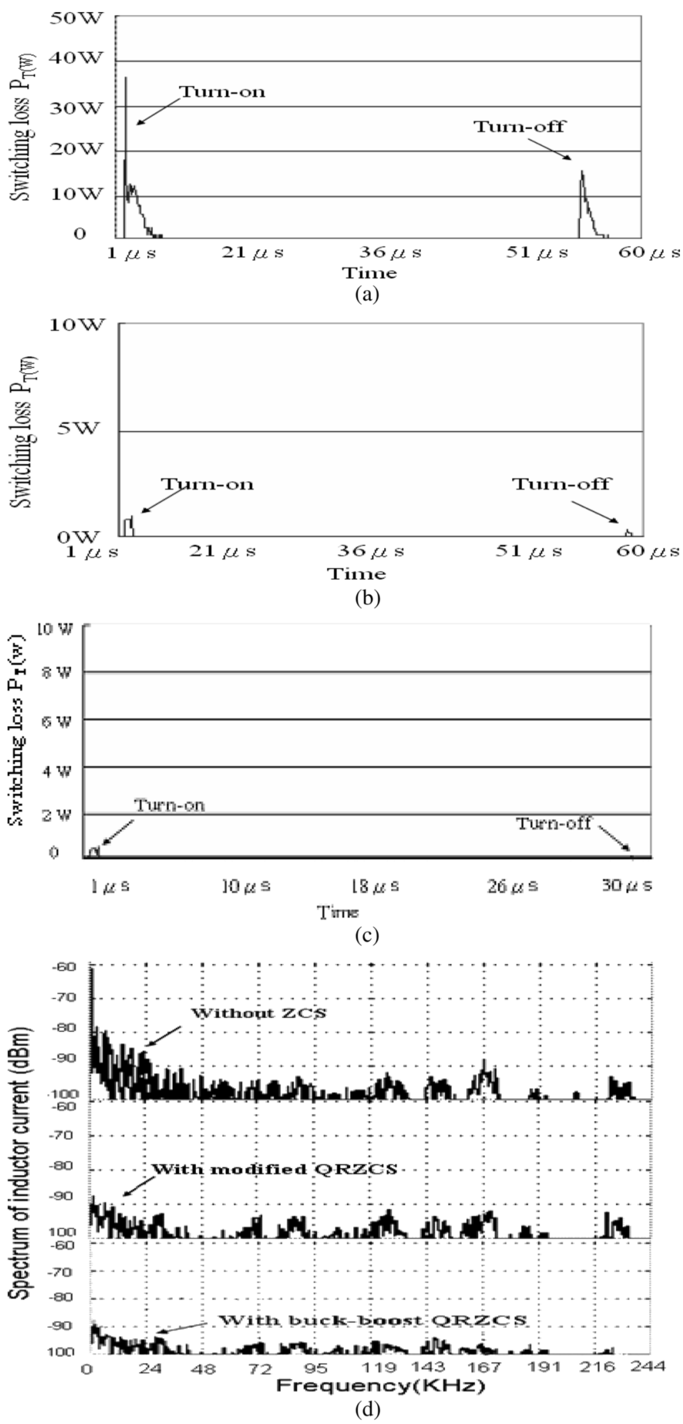

Fig. 13. Experimental results of the MOSFET switch power losses for various equalizers. (a) Experimental switch power losses for the equalizer without QRZCS. (b) Experimental switch power losses for the equalizer with modified QRZCS. (c) Experimental switch power losses for the equalizer with buck–boost QRZCS. (d) Experimental FFT spectrum for the equalizer without and with QRZCS.

the buck–boost QRZCS battery equalizer. This makes it suitable for low cost modular design and low voltage lithium–ion battery equalization applications.

- The effort to increase the operating frequency, reduce the weight, size, and cost of the magnetic and filter elements is an additional benefit of the proposed QRZCS battery

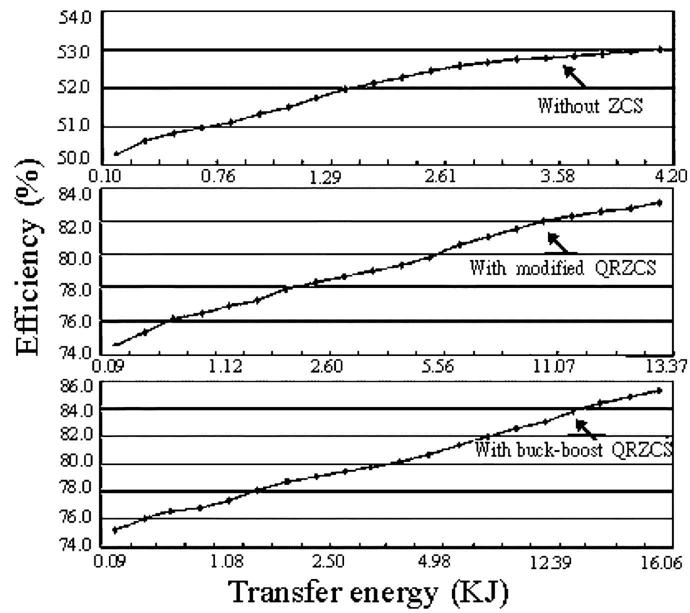

Fig. 14. Experimental energy transfer efficiency for the various equalizers during battery string equalization.

equalization technology. The power MOSFET switches of the modified QRZCS converter are not turned on exactly at the zero-current state, which increases the switching losses and decreases the equalization efficiency, making the modified QRZCS less efficient than the buck–boost QRZCS battery equalizer. This disadvantage can be improved using a fully soft-switching extended-period quasi-resonant converter [17] or an inverting type ZCS switched-capacitor bidirectional converter [18], or the concept of the zero-current/zero-voltage switching bidirectional converter technology [19], [20].

- The values of the resonant inductance L_r and capacitance C_r are effected by many desired specifications, such as maximum switching frequency, cell voltages and their variations, equalization currents and their variations, normalized impedance, ..., etc. The design guidelines and the effects caused by the values of the L_r and C_r for the ZCS-quasi-resonant-converters are in detail derived from and proposed in the literature [16]. The effects in the modified QRZCS battery equalizer will be subjected to later study.

VI. Conclusion

The cell balancing control schemes based on the QRZCS converter theory for the bidirectional cell equalization of a series connected battery strings have been studied. It is found that these cell balancing control schemes possess interesting properties such as reducing the switching loss, increasing the battery equalization efficiency, and significantly extending the battery life and battery string capability. Simulations and experimental results have been performed to verify the theoretical analysis. A comparison has also been made between ICE without and with QRZCS operations for lithium–ion battery equalization. It is shown that the modified QRZCS battery equalizer is more suitable for low cost and high efficiency battery pack applications.

REFERENCES

[1] S. T. Tung, D. C. Hopkins, and C. R. Mosling, "Extension of battery life via charge equalization," *IEEE Trans. Ind. Electron.*, vol. 40, no. 1, pp. 96–104, Feb. 1993.

[2] ——, "The use of equalizing converter for serial charging of long battery strings," in *Proc. 6th Annu. IEEE Appl. Power Electron. Conf. Expo (APEC'91)*, 1991, pp. 493–498.

[3] N. H. Kutkut, D. M. Divan, and D. W. Novotny, "Charge equalization for series connected battery strings," in *Proc. IEEE IAS Annu. Meeting*, Oct. 1994, pp. 1008–1015.

[4] N. H. Kutkut and D. M. Divan, "Dynamic equalization techniques for series battery stacks," in *Proc. 18th Int. Telecommun. Energy Conf. (INTELEC'96)*, 1996, pp. 514–521.

[5] N. H. Kutkut, "Non-dissipative current diverter using a centralized multi-winding transformer," in *Proc. 28th Annu. IEEE Power Electron. Spec. Conf. (PESC'97)*, 1997, vol. 1, pp. 648–654.

[6] W. F. Bentley, "Cell balancing considerations for lithium–ion battery systems," in *Proc. 12th Annu. Battery Conf. Appl. Adv.*, Jun. 14–17, 1997, pp. 223–226.

[7] S. W. Moore and P. J. Schneider, "A review of cell equalization methods for lithium ion and lithium polymer battery systems," *SAE Tech. Paper Series*, pp. 1–5, Mar. 2001.

[8] N. H. Kutkut, "A modular nondissipative current diverter for EV battery charge equalization," in *Proc. 13th Annu. IEEE Appl. Power Electron. Conf. Expo (APEC'98)*, 1998, vol. 2, pp. 686–690.

[9] Y. S. Lee and M. W. Cheng, "Intelligent control battery equalization for series connected lithium–ion battery strings," *IEEE Trans. Ind. Electron.*, vol. 52, no. 5, pp. 1297–1307, Oct. 2005.

[10] Y. S. Lee and C. W. Jao, "Fuzzy controlled lithium–ion battery equalization with state-of-charge estimators," in *Proc. IEEE SMC'03 Conf.*, 2003, pp. 2024–2045.

[11] H. Chung, S. Y. R. Hui, and K. K. Tse, "Reduction of power converter EMI emission using soft-switching technique," *IEEE Trans. Electromagn. Compat.*, vol. 40, no. 3, pp. 282–287, Aug. 1998.

[12] Y. Tang, H. Zhu, B. Song, J. S. Lai, and C. Chen, "EMI experimental comparison of PWM inverter between hard- and soft-switching techniques," *Power Electron. Transport.*, pp. 71–77, 1998.

[13] K. H. Liu and F. C. Lee, "Zero-current switching technique in DC/DC converter circuits," in *Proc. IEEE Power Electron. Spec. Conf. (PESC'86)*, 1986, pp. 58–70.

[14] K. H. Liu, R. Oruganti, and F. C. Lee, "Quasi-resonant converter-topologies and characteristics," *IEEE Trans. Power Electron.*, vol. PE-2, no. 1, pp. 62–71, Mar. 1987.

[15] F. L. Luo and Y. Hong, "Four-quadrant DC/DC zero-current-switching quasi-resonant Luo-converter," in *Proc. 32nd Annu. IEEE Power Electron. Spec. Conf. (PESC'01)*, 2001, vol. 2, pp. 878–883.

[16] A. W. Lotfi, V. Vorperian, and F. C. Lee, "Comparison of stresses in quasi-resonant and pulse-width-modulation converters," in *Proc. IEEE Power Electron. Spec. Conf. (PESC'88)*, 1988, pp. 591–598.

[17] S. Y. R. Hui, K. W. E. Cheng, and S. R. N. Prakash, "A fully soft-switching extened -period quasi-resonant power-factor-correction circuit," *IEEE Trans. Power Electron.*, vol. 12, no. 5, pp. 922–930, Sep. 1997.

[18] Y.-S. Lee and Y.-Y. Chiu, "Zero current switching switched-capacitor bidirectional DC–DC converter," *Proc. Inst. Elect. Eng.*, vol. 152, no. 6, pp. 1525–1530, Nov. 2005.

[19] Y. Ren, M. Xu, C. S. Leu, and F. C. Lee, "A family of high power density bus converters," in *Proc. IEEE Power Electron. Spec. Conf. (PESC'04)*, Aachen, Germany, 2004, pp. 527–532.

[20] B. Ray, "Bidirectional DC/DC power conversion using constant-frequency quasi-resonant topology," in *Proc. IEEE Int. Symp. Circuits Syst. (ISCAS'93)*, 1993, pp. 2347–2350.

Yuang-Shung Lee (M'91) received the M.Sc. and Ph.D. degrees in electrical engineering from the National Taiwan Institute of Technology, Taipei, Taiwan, R.O.C., in 1983 and 1993, respectively.

In 1986, he joined Fu-Jen Catholic University, Taipei, Taiwan, R.O.C., as a faculty member and is presently a Professor. He was Head of the Department of Electronic Engineering from 1997 to 2000. His current research interests are charge equalization and protection of lithium–ion battery systems, soft switching power converter design for fuel cell systems, EMI measurement and filter design of power supply, fuzzy neural network application and control, and powerline communication for digital homes.

Guo-Tian Cheng received the M.Sc. degree from the Institute of Electronic Engineering, Fu-Jen Catholic University, Taipei, Taiwan, R.O.C., in 2004.

In November 2004, he joined the Energy and Resource Laboratory, Industrial Technology Research Institute, Hsinchu, Taiwan, where he is currently a Design Engineer for switching mode power supplies and fuel cell power converters. His research interests include soft switching power converter designs and fuel cell energy conversion systems.